Total No. of Questions : 4]

SEAT No. :

PE-531

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S.E. (Civil) (Insem.)

ENGINEERING MATHEMATICS - III

(2019 Pattern) (Semester - III) (207001)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Answer Q.1 or Q.2, Q.3 or Q.4.*
2) *Use of electronic pocket calculator is allowed.*
3) *Assume suitable data if necessary.*
4) *Neat diagrams must be drawn wherever necessary.*
5) *Figures to the right indicate full marks.*

Q1) a) Solve the following differential equations (Any two) :

i) $(D^2 - 4D + 4)y = e^x + \sin 2x + x$ **[5]**

ii) $(D^2 + 1)y = \cot x$ [Use variation of parameter method) **[5]**

iii) $(x+1)^2 \dfrac{d^2y}{dx^2} + (x+1)\dfrac{dy}{dx} + y = \cos[\log(1+x)]$ **[5]**

b) The deflection of a strut of length 'l' with one end ($x = 0$) built-in and the other supported, subjected to end thrust 'p' satisfies the differential equation. **[5]**

$\dfrac{d^2y}{dx^2} + a^2y = \dfrac{a^2}{P}R(l-x)$, prove that the deflection curve is $y = \dfrac{R}{P}$ $\left(\dfrac{\sin ax}{a} - l\cos ax + l - x\right)$ where $al = \tan al$

OR

Q2) a) Solve the following differential equations (Any two) :

i) $\dfrac{d^2y}{dx^2} + 4\dfrac{dy}{dx} + 4y = e^{-2x}\sec^2 x(1 + 2\tan x)$ **[5]**

ii) $\dfrac{d^2y}{dx^2} + 4y = x\sin x$ **[5]**

iii) $\dfrac{dx}{3z-4y} = \dfrac{dy}{4x-2z} = \dfrac{dz}{2y-3x}$ **[5]**

P.T.O.

b) A horizontal beam is uniformly loaded. It's one end fixed and other end is subjected to a tensile force 'P'. The deflection of the beam is given by $EI\frac{d^2y}{dx^2} = Py - \frac{Wx^2}{2}$ given that $\frac{dy}{dx}=0$, $y = 0$ at $x = 0$ show that the deflection of a beam for a given x is

$$y = -\frac{W}{2Pn^2}\left[e^{nx} + e^{-nx} - n^2x^2 - 2\right]$$

where $n^2 = \frac{P}{EI}$ **[5]**

Q3) a) Use Gauss-elimination method with Partial Pivoting to solve following system of equations **[5]**

$8y + 2z = -7$

$3x + 5y + 2z = 8$

$6x + 2y + 8z = 26$

b) Use the Runge-Kutta fourth order method to solve **[5]**

$\frac{dy}{dx} = x + y$, $y(1) = 1.5$ at $x = 1.1$ with h = 0.1

c) Solve following system of equations by using Cholesky-method. **[5]**

$4x_1 + 6x_2 + 8x_3 = 0$

$6x_1 + 34x_2 + 52x_3 = -160$

$8x_1 + 52x_2 + 129x_3 = -452$

OR

Q4) a) Apply Gauss-Seidel method to solve the equations. **[5]**

$27x_1 + 6x_2 - x_3 = 85$

$6x_1 + 15x_2 + 2x_3 = 72$

$x_1 + x_2 + 54x_3 = 110$

b) Using modified Euler's method find an approximate value of y when $x = 0.1$ given that $\frac{dy}{dx} = x + y$; $y(0) = 1$, by taking $h = 0.1$ **[5]**

c) Numerical Solution of the differential Equation $\frac{dy}{dx} = \frac{2 - y^2}{5x}$ is tabulated as **[5]**

x	4	4.1	4.2	4.3
y	1.0	1.0049	1.0097	1.0143

Use Milne's predictor-Corrector method to find y at $x = 4.4$ by taking h = 0.1